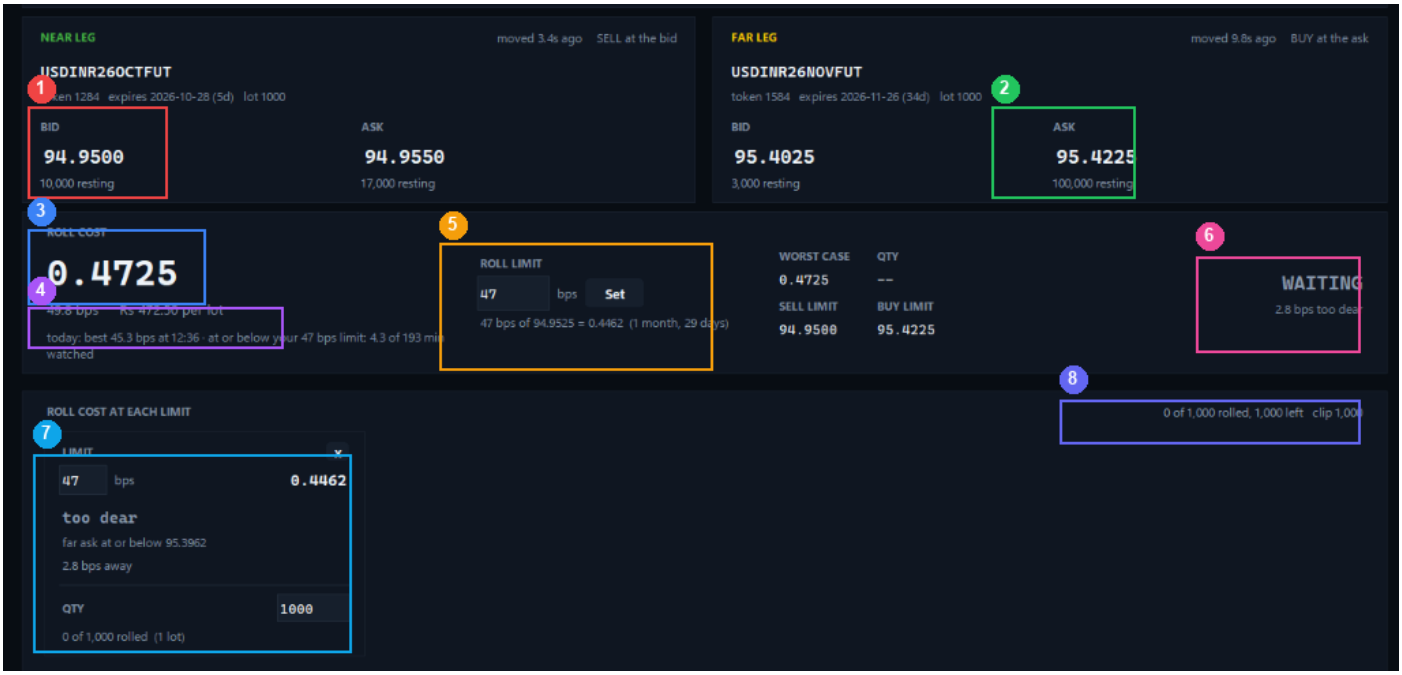
USD/INR Rollover Watcher & Execution Engine

Visual Screen Map & Mathematical Calculation Reference (Step-by-Step)



1. Live Screen Breakdown (Matching the Numbered Callout Pins Above)

[1] Near Leg Bid (94.9500) **Action: SELL October Futures** To exit your expiring Oct contract, the order hits the live Bid. You receive **94.9500 Rs** per USD. *Resting depth: 10,000 qty.*	**[2] Far Leg Ask (95.4225)** **Action: BUY November Futures** To enter the forward Nov contract, the order lifts the live Ask. You pay **95.4225 Rs** per USD. *Resting depth: 100,000 qty.*

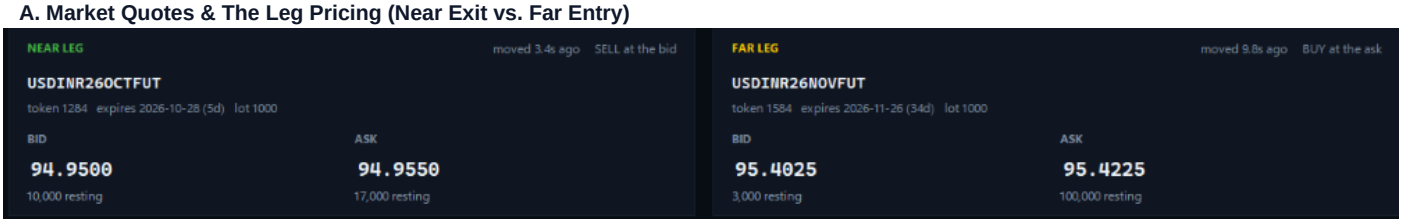
[3] Roll Cost Spread (0.4725 Rs)
`Formula: Far Ask - Near Bid`
`95.4225 - 94.9500 = 0.4725 Rs`
`The exact difference in rupees per dollar paid to carry the hedge 1 month forward.`
	[4] Basis Points & Cost / Lot
	`Formula: (Spread / Mid) x 10,000`
	`Mid = (94.9500 + 94.9550)/2 = 94.9525`
	`(0.4725 / 94.9525) x 10,000 = 49.8 bps`
	`Per Lot: 0.4725 x 1,000 = Rs 472.50 per lot`

[5] Roll Limit Ceiling (47 bps = 0.4462)
`Your Configured Ceiling`
`You configured 47 bps for 1-month rolls.`
`(47 / 10,000) x 94.9525 = 0.4462 Rs`
`The maximum allowable spread. The bot will reject anything higher.`
	[6] Live Status: WAITING (2.8 bps too dear)
	Decision Rule: Cost <= Limit
	Market Cost = 49.8 bps | Your Limit = 47.0 bps
	49.8 - 47.0 = 2.8 bps too dear (too expensive).
	The system holds safely without executing until market price qualifies.

[7] Target Trigger Hint (95.3962) **When will it execute?** Near Bid + Limit = 94.9500 + 0.4462 = 95.3962 If Nov Ask falls to <= **95.3962** (or Oct Bid rises by 2.8 bps), the trade fires automatically.	**[8] Sizing & Progress (1,000 left, clip 1,000)** **Trade Execution Sizing** • **1 Lot** = **1,000 Quantity**. • **Each Clip:** Sends **1,000 qty in one go** (SELL 1,000 Oct / BUY 1,000 Nov). • Capped at 1 clip per day for safety.

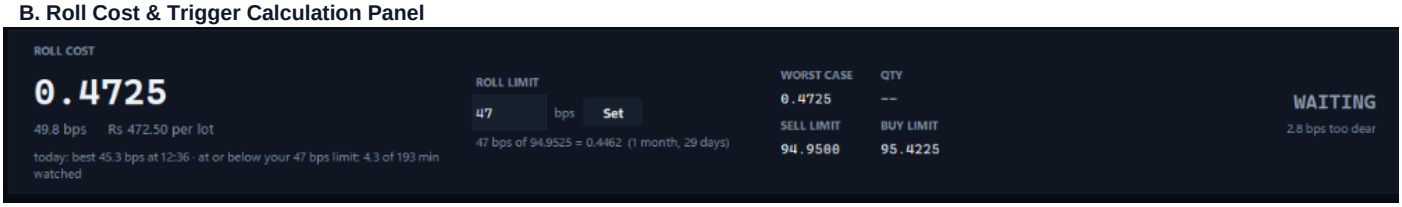
2. Deep Dive: Section-by-Section Screen Walkthrough

A. Market Quotes & The Leg Pricing (Near Exit vs. Far Entry)



- **Near Leg (Oct FUT):** You are closing a long position, so the system places a SELL limit at the **Near Bid (94.9500)**.
- **Far Leg (Nov FUT):** You are opening the new forward contract, so the system places a BUY limit at the **Far Ask (95.4225)**.
- **Execution Safeguard:** Both limit orders are sent at current touchline prices with allowance_ticks = 0 to guarantee zero negative price slippage.

B. Roll Cost & Trigger Calculation Panel



- **Observed Spread:** 95.4225 - 94.9500 = 0.4725 Rs (= **49.8 bps** / **Rs 472.50 per lot**).
- **Roll Limit:** Set to **47 bps** (= **0.4462 Rs**). The mathematical rule is: **Trade ONLY if Spread <= Limit**.
- **Waiting Indicator:** Because 49.8 bps is **2.8 bps higher** than 47.0 bps, the system displays WAITING: 2.8 bps too dear and holds safely.
- *Historical Tracker:* Shows today's best roll cost was **45.3 bps** (at 12:36 PM), and the market has been at/below your limit for 4.3 minutes today.

3. Sizing Rules: From 4,000 Lots Total to 1 Lot (1,000 Qty) Per Trade

HOW THE BOT SIZES EACH CLIP

- **Total Account Position: 4,000 Lots** = **4,000,000 Quantity** (40 Lakh USD) across October futures.
- **Single Trade Size:** Configured to lots: 1. When the trade triggers, it executes **exactly 1,000 quantity in one go**.
- **Why 1,000 Quantity?** Choice Broking exchange API strictly requires orders in base units (1 lot = 1,000 units). Sending '1' would be rejected.
- **Automatic Daily Lock:** max_clips_per_day: 1 guarantees that after this 1 lot rolls, the engine immediately locks and prevents further trades today.

4. Operator 4-Step Checklist (How to Execute via GUI)

Step	Action in Application Window (dist/roll_app.exe)	Expected Screen Confirmation
Step 1	**Open roll_app.exe**: Inspect the live Near Bid (94.9500) and Far Ask (95.4225).	Live prices and current roll cost (0.4725 / 49.8 bps) tick in real time.
Step 2	**Set Target Limit**: Enter target bps (e.g. 47 bps to wait, or 50 bps to trigger immediately) and click **Set**.	The limit card updates immediately with the new rupee threshold (e.g. 0.4462 or 0.4748).
Step 3	**Switch from DRY RUN to LIVE**: Click the **DRY RUN** button at top right. A safety dialog pops up. Type LIVE and confirm.	Top status bar turns into a bright red **LIVE** badge. Real exchange orders are now authorised.
Step 4	**Click ARM**: System enters armed state for up to 120 seconds.	When Cost <= Limit, bot sends 1,000 Oct Sell and 1,000 Nov Buy in one go, fills cleanly, and halts.